HAND & UPPER LIMB · NEET-SS / FET / INI-SS

flexor and extensor tendons

Flexor tendons

- **Zones (Verdan):**
  - I: distal to the FDS insertion (FDP only).
  - **II: "no man's land"**, from the A1 pulley to the FDS insertion (FDS and FDP in one sheath).
  - III: palm. IV: carpal tunnel. V: forearm.
  - Thumb: T1-T3.
- **Pulleys:** annular A1, A3 and A5 lie over joints. **A2 and A4 are the most important**: losing them causes bowstringing.
- **Repair principles:**
  - Repair strength rises with the **number of core strands** crossing the repair: use 4 or more strands plus an epitendinous suture.
  - Repair within about 2 weeks.
  - Start **early active motion** protocols.
  - Repairs are weakest at about **days 6-12**, the classic rupture window.
- **FDP avulsion (Jersey finger): ring finger** in about 75%. **Leddy-Packer:**
  - Type I: retracted to the palm, both vincula torn; repair within 7-10 days.
  - Type II: held at the PIP level by the long vinculum; repair possible up to about 6 weeks.
  - Type III: large bony fragment caught at A4; ORIF.
  - Type IV: bony fragment, with the tendon also avulsed from it.

- **Quadriga effect:** FDP tendons share a muscle belly, so advancing or tethering one FDP limits flexion of the neighbours.
- **Lumbrical-plus finger:** FDP divided distal to the lumbrical origin; trying to flex the finger extends the IP joints paradoxically.
- **Trigger finger:** A1 pulley; steroid injection first (works less well in diabetics); open or percutaneous A1 release.

Extensor tendons

- **Zones I-VIII:** odd zones lie over joints (I DIP, III PIP, V MCP, VII wrist). Thumb: T I-V.
- **Mallet finger (zone I):**
  - Splint the DIP in extension **continuously for 6-8 weeks**, then at night.
  - Surgery for **volar subluxation of the distal phalanx**; large articular fragments are debated.
  - An untreated mallet can lead to **swan-neck deformity**.
- **Boutonniere (zone III):**
  - Central slip rupture lets the lateral bands slip volar to the PIP axis.
  - The **Elson test** detects it early, before the deformity is fixed.
  - Splint the **PIP in extension for 6 weeks with the DIP free** to flex (keeps the lateral bands mobile).
- **Sagittal band rupture (boxer's knuckle):** most often the middle finger, radial band; the extensor tendon subluxes ulnarward.
- **EPL rupture after an undisplaced distal radius fracture:** attritional, at Lister's tubercle (watershed area); treat with **EIP-to-EPL transfer**.

Tenosynovitis around the wrist

- **De Quervain:** first dorsal compartment (**APL and EPB**). Finkelstein (or Eichhoff) test. An EPB in a separate subsheath can explain a failed injection. Protect the **superficial radial nerve** at release.
- **Intersection syndrome:** where APL/EPB cross ECRL/ECRB (second compartment), about 4-6 cm proximal to Lister's tubercle; crepitus in rowers.
- **Dorsal compartments:** 1 APL/EPB; 2 ECRL/ECRB; 3 EPL; 4 EDC/EIP (with the PIN); 5 EDM; 6 ECU.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.